STUDENT PERFORMANCE ANALYSIS AND PREDICTION SYSTEM

A Python and Machine Learning Project Report

Prepared using the five visualizations and evaluation values supplied for this project

Project focus

Exploratory analysis of student study habits and examination scores, followed by a Linear Regression model for examination-score prediction.

Note: Add the student's name, institution, course, training-program details, dataset source, and exact dataset size before final submission. These details were not assumed.

1. Introduction

Student academic performance can be explored alongside factors such as study time, sleep, attendance, and previous scores. This project uses Python for data analysis and visualization, then applies Linear Regression to estimate the examination-score target. The analysis is intended to identify patterns in the supplied dataset and demonstrate a basic end-to-end machine-learning workflow.

2. Objectives

• Inspect and summarize the student-performance variables.

• Use visualizations to explore score distributions and relationships.

• Examine correlations between selected numerical features and examination scores.

• Train a Linear Regression model using student-related input features.

• Evaluate model predictions using MAE, RMSE, and R^2.

3. Dataset and Variables

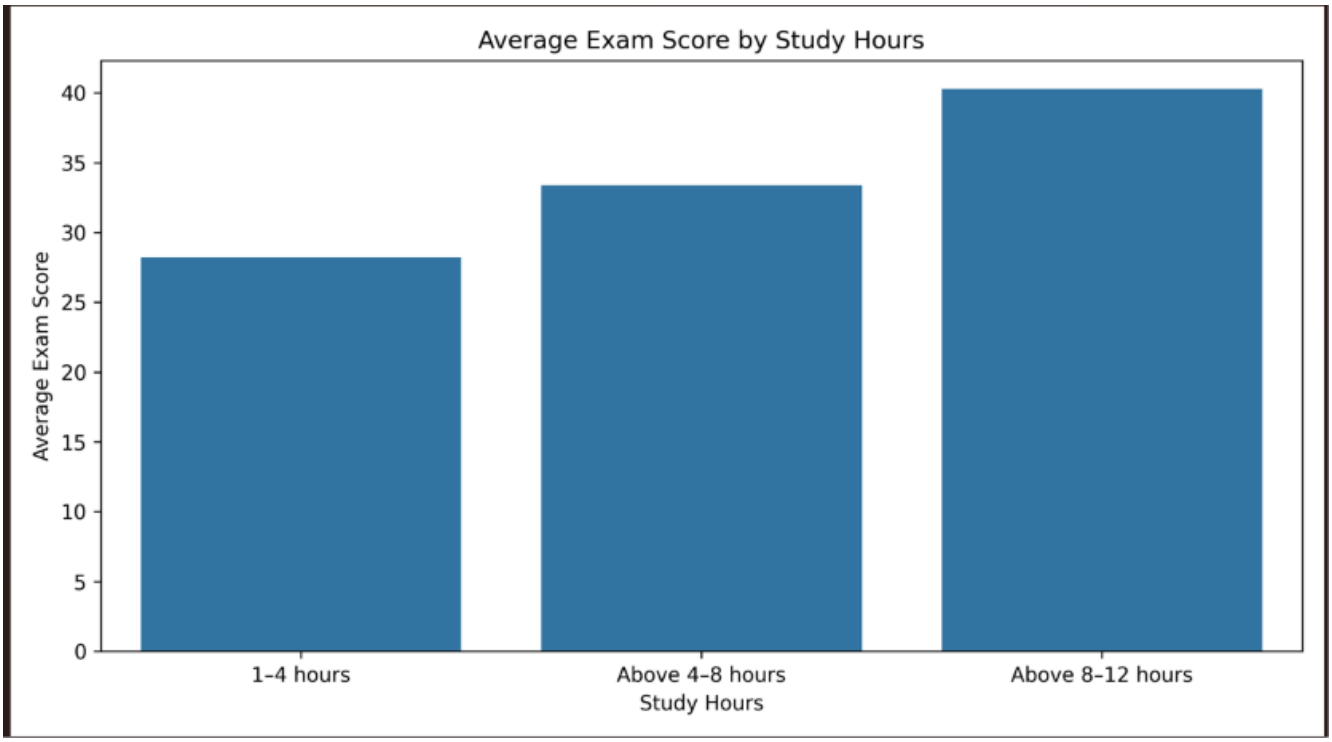
The supplied visualizations show the following variables: **hours_studied**, **sleep_hours**, **attendance_percent**, **previous_scores**, and **exam_score**. The prediction target is **exam_score**. The four other variables appear as candidate explanatory features in the correlation heatmap.

Dataset provenance: *Insert the exact dataset name, source URL or attribution, record count, and any preprocessing performed, as recorded in the original notebook. These details are left for verification rather than being invented.*

4. Exploratory Data Analysis and Visualizations

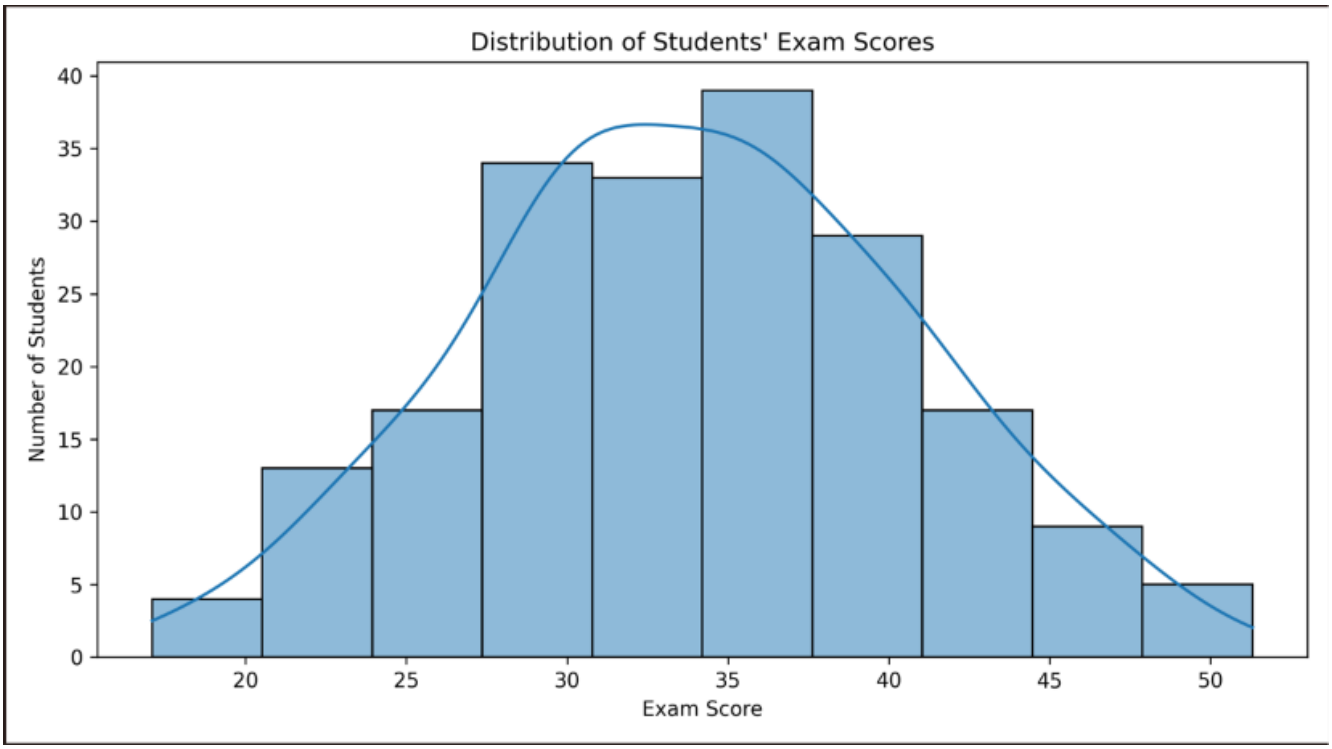
Five supplied graphs are included below. Their interpretations describe visible patterns in the charts and should not be treated as evidence of causation.

Figure 1. Average Exam Score by Study Hours



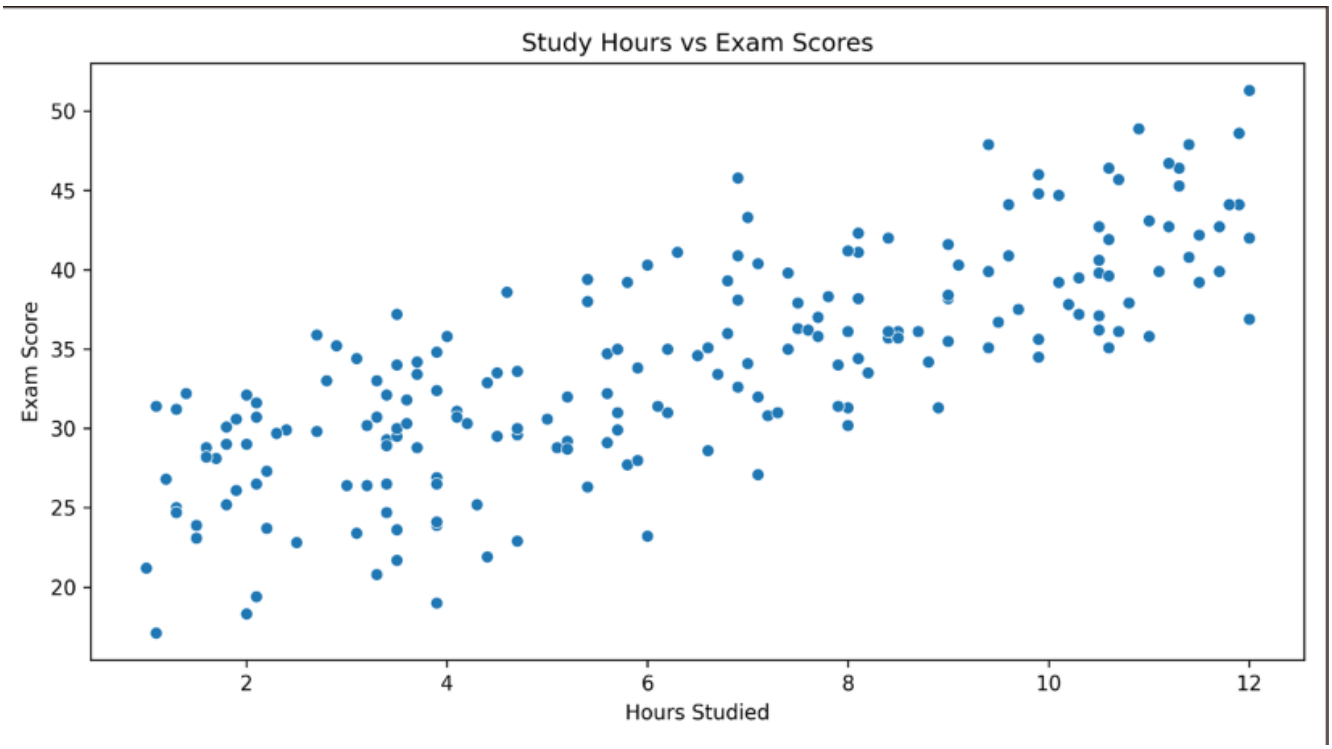
The grouped bar chart shows a rising average exam score across the displayed study-hour categories. The highest study-hour group has the highest displayed average.

Figure 2. Distribution of Students' Exam Scores



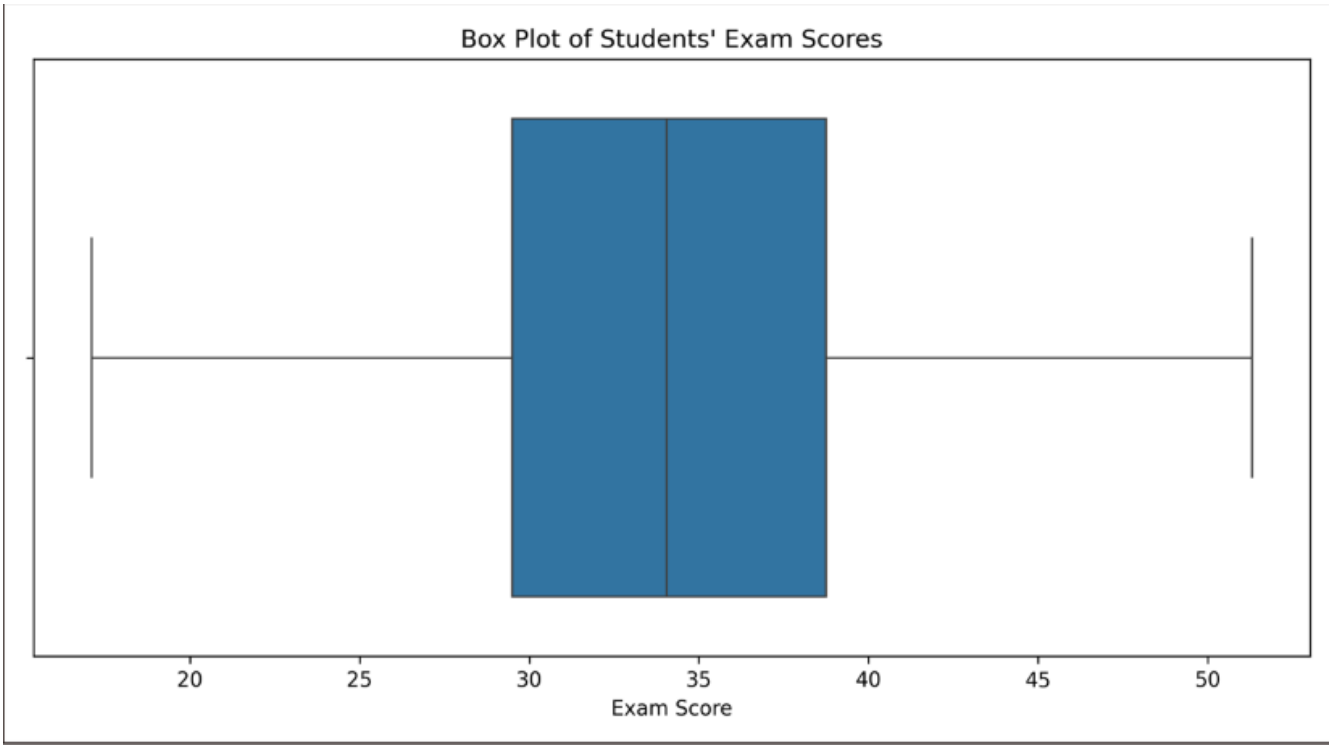
The histogram shows the distribution of examination scores. Most observations in the supplied chart appear concentrated around the low-to-high 30s, with fewer observations toward the tails.

Figure 3. Study Hours vs Exam Scores



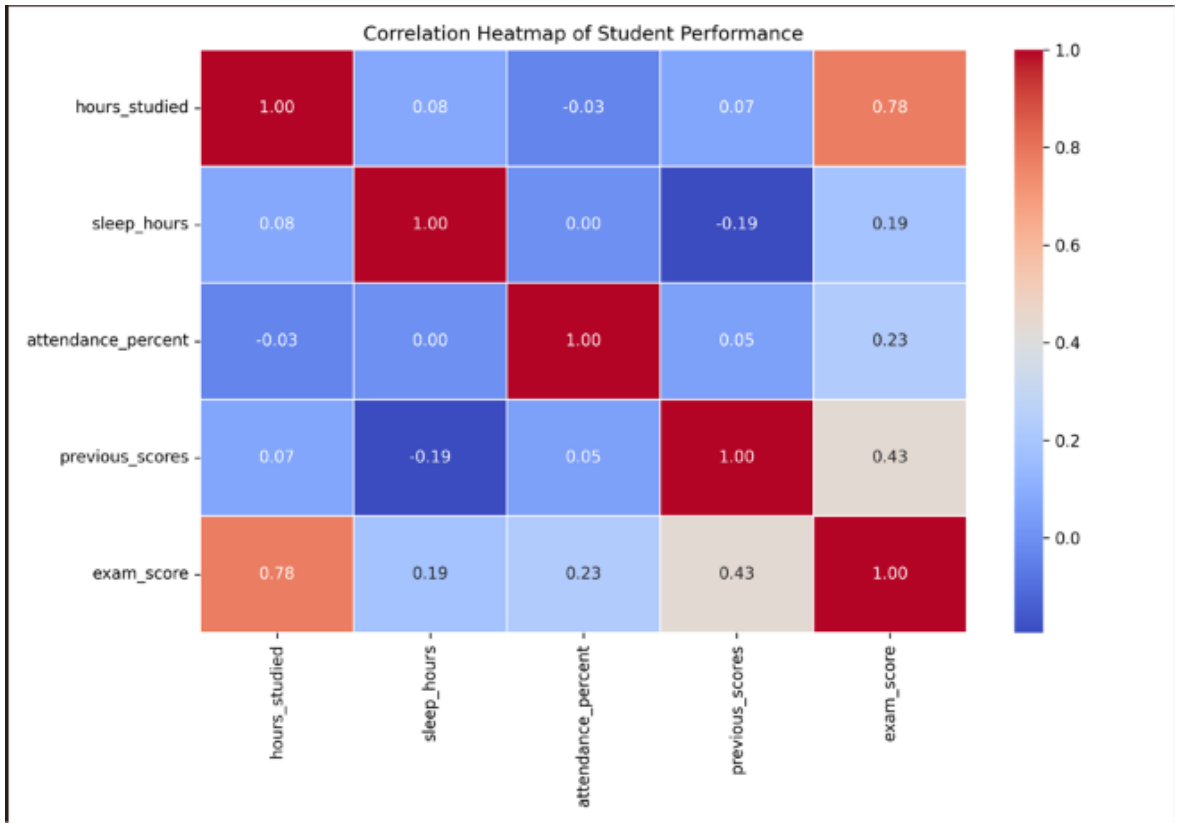
The scatter plot shows a generally positive association: higher study hours tend to appear alongside higher examination scores. There is still variation among students at similar study-hour values.

Figure 4. Box Plot of Students' Exam Scores



The box plot summarizes the median, interquartile range, and spread of examination scores. No individual outlier markers are visibly plotted in the supplied figure.

Figure 5. Correlation Heatmap of Student Performance



The displayed correlation between hours_studied and exam_score is 0.78; previous_scores and exam_score is 0.43; attendance_percent and exam_score is 0.23; and sleep_hours and exam_score is 0.19. These values describe linear association in this dataset, not causal effects.

5. Machine Learning Methodology

Algorithm: Linear Regression.
Prediction target: exam_score.
Candidate input features: hours_studied, sleep_hours, attendance_percent, and previous_scores.

The project workflow previously discussed used an 80% training and 20% testing split. Confirm that this split and the feature list match the final notebook before submission. The model is trained on the training data and evaluated on held-out test data.

6. Model Evaluation

The following values are the model evaluation results previously supplied by the project owner. They are presented as the held-out test metrics; verify that they match the final notebook output.

Metric	Value	Interpretation
MAE	2.31085823589583	Average absolute prediction error, in exam-score units.
RMSE	2.78599760743312	Penalizes larger errors more strongly than MAE.
R^2 Score	0.853744713715951	Proportion of target variance explained on the evaluated test set.

The R^2 score is not an accuracy percentage. MAE and RMSE are measured in the same units as the target score. Interpretation of whether an error is small or large depends on the score scale and context.

7. Limitations

• The analysis reflects only the dataset used in the notebook; its size and representativeness must be confirmed.

• Associations in the charts do not prove that a study habit causes a change in exam scores.

• The model's performance may differ on new students or data collected in another setting.

• Cross-validation results should only be reported after the cross-validation code has been run and its actual output recorded.

8. Conclusion

The supplied visualizations show a positive association between study hours and examination scores, and the correlation heatmap also shows positive correlations between exam_score and previous_scores, attendance_percent, and sleep_hours. The Linear Regression model's reported test metrics are MAE 2.3109, RMSE 2.7860, and R^2 0.8537. These results provide an initial demonstration of student-performance analysis and prediction. They should be interpreted within the limits of the dataset and verified against the final executable notebook.

9. Items to Complete Before Submission

■ Add student name, institution, course, and training-program details.

■ Insert the exact dataset title, source/attribution, and record count from the original dataset.

■ Verify the feature columns, train/test split, and evaluation values against the final notebook.

■ Add cross-validation results only if successfully generated.

■ Ensure the notebook, dataset, requirements.txt, README.md, report, and any generated model/output files are included in the repository.